Elements of AIML
Unit 05 – Lecture 03 Notes
Applications in Education, Insurance, Retail, Transport & Energy (Wrap-up)

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1 Lecture Overview

This final lecture completes Unit V by surveying additional application domains and providing:

- a practical deployment checklist (what can go wrong),

- a revision set that connects Units I–V.

Sources: NIST, *AI Risk Management Framework (AI RMF 1.0)*, 2023.;Barocas, Hardt, and Narayanan, *Fairness and Machine Learning* (online book), selected sections.;Knaflic, *Storytelling with Data*, selected chapters on communicating results.

2 How to use these notes (self-study)

Use this lecture as an integration lesson. Pick one domain (education/insurance/retail/transport/energy) and apply the checklist: define the task, pick metrics, identify risks, and propose a monitoring plan. Then use the revision questions to connect logic (Unit II) with evaluation (Unit III) and modeling (Unit IV).

3 Learning Outcomes

- List use-cases in education, insurance, retail/supply chain, transport, and energy.
- Explain domain constraints (metrics, drift, safety, privacy).
- Apply a checklist to evaluate deployment risks.
- Answer revision questions across the course.

4 Keywords

- **Drift:** changing patterns over time.
- **Feedback loop:** model outputs influence future data (recommendations).
- **Checklist:** structured risk review for deployment.

5 Abbreviations and key terms

- **KPI**: key performance indicator (business success measure).
- **ETA**: estimated time of arrival (transport).
- **SLA**: service-level agreement (latency/uptime requirements).
- **Objective mismatch**: optimizing the wrong metric for the real-world goal.

6 Applications across domains

6.1 Education

- personalization and recommendation,
- early warning for dropout risk,
- analysis of learning outcomes.

Important constraints: privacy, fairness, transparency.

Typical data/labels (examples). Learning activity logs, assessment scores, attendance, and engagement signals. Labels could be "dropout" or "course completion". Metrics must consider fairness (avoid disadvantaging groups) and privacy of student data.

6.2 Insurance

- risk modeling and pricing support,
- claim triage and fraud detection,
- customer support automation.

Important constraints: explainability, fairness, regulation.

Typical tasks and risks. Fraud detection is imbalanced; pricing models require transparency and governance. Proxy features can create unfair outcomes, and models must support audit/appeals.

6.3 Retail and supply chain

- demand forecasting and inventory management,
- recommendation systems,
- logistics optimization.

Important constraints: seasonality/drift, feedback loops, data quality.

Feedback loop example. If recommendations increase exposure of some products, the data collected reflects the system's own choices, which can amplify popularity bias unless corrected.

6.4 Transportation and logistics

- ETA prediction, routing,
- predictive maintenance,
- safety monitoring.

Important constraints: real-time requirements, safety, sensor errors.

Metrics. ETA uses MAE/RMSE (minutes). Routing/dispatch may optimize cost/time subject to safety and SLAs. Sensor failures and drift require monitoring and fallback strategies.

6.5 Energy and utilities

- load forecasting,
- fault detection,
- demand-response optimization.

Important constraints: reliability, rare events, monitoring.

Rare-event issue. Fault detection resembles imbalanced classification/anomaly detection: most time is normal operation, so accuracy can be misleading.

7 What could go wrong? (deployment checklist)

Use this checklist before and after deployment:

1. **Objective mismatch:** are we optimizing the right metric/cost?
2. **Data problems:** leakage, bias, missing values, non-representative data.
3. **Fairness/privacy:** are there subgroup gaps? are we compliant?
4. **Drift:** what changes over time? what is the monitoring plan?
5. **Security/misuse:** adversarial inputs, abuse, unauthorized access.

8 Common pitfalls (course wrap-up)

- **Thinking "accuracy is everything":** real systems require the right metric, fairness checks, and cost-aware decisions.
- **Ignoring feedback loops:** recommendations and policies change user behavior, which changes the future data.
- **No monitoring/retraining plan:** drift is normal; lack of monitoring turns small issues into failures.
- **Skipping documentation:** without documentation, auditing and debugging deployed models becomes very difficult.

9 Practice (with answers)

P1. Give one KPI for an education recommender system.

Answer: improvement in course completion rate, or increased engagement time, while ensuring fairness across groups.

P2. Why are time-based splits important for forecasting tasks?

Answer: the model will be used on future data; random shuffling can leak future patterns into training and produce overly optimistic estimates.

10 Revision questions (with answers)

R1. What is entailment and how can we test it using refutation?

Answer: entailment $KB \models \alpha$ means α is true in every model of KB. Refutation test: check if $KB \wedge \neg\alpha$ is unsatisfiable (derive contradiction via resolution).

R2. Why do we need train/validation/test splits?

Answer: to measure generalization honestly. Train fits parameters, validation tunes choices, test gives an unbiased final evaluation. Reusing test for tuning causes optimistic bias.

R3. Regression vs classification (and metrics)?

Answer: regression predicts continuous values (MAE/RMSE/R^2). Classification predicts labels (precision/recall/F1/ROC-AUC). Metric choice depends on error costs and imbalance.

R4. What does PCA optimize and why standardize first?

Answer: PCA finds orthogonal directions of maximum variance. Standardization prevents large-scale features from dominating the covariance and components.

R5. Give two examples of drift and one monitoring method.

Answer: fraud patterns change; customer behavior changes seasonally; spam tactics evolve. Monitoring: track feature distributions and performance metrics over time with alerts.

R6. What is class imbalance and why is accuracy misleading?

Answer: imbalance means one class is rare. Accuracy can be high by predicting the majority class; use precision/recall/F1 and cost-aware evaluation.

R7. What is unification used for in logic inference?

Answer: unification finds substitutions to match FOL literals so resolution can be applied with variables (e.g., $Human(x)$ with $Human(Socrates)$).

R8. Give one example of a feedback loop in recommendations.

Answer: if a recommender shows certain products more, users click them more, which produces data that further reinforces recommending the same products, potentially reducing diversity and increasing bias.

R9. Why is explainability important in insurance?

Answer: decisions affect people financially; regulations and trust require understandable reasons; explainability supports auditing and contesting errors.

R10. What are two responsible AI checks before deployment?

Answer: fairness evaluation across subgroups and privacy/security review (data governance, access control), plus monitoring plan for drift.

11 Exit questions (with answers)

Q1. Pick one domain and propose an ML task + suitable metric.

Answer: retail demand forecasting (regression) with RMSE/MAE; fraud detection (classification) with precision/recall; healthcare screening with high recall.

Q2. List three checklist items.

Answer: objective mismatch, data leakage/bias, drift and monitoring, fairness/privacy, security/misuse (any three).

Q3. Why explainability in insurance?

Answer: accountability and regulation; decisions must be justified and audited.

Q4. One example of feedback loop in retail recommendations.

Answer: popular items get more exposure, then more clicks, then even more exposure, causing self-reinforcing popularity bias.

12 Summary

This course combined logic-based reasoning and data-driven learning, with emphasis on evaluation and responsible application in real domains.

Sources: NIST, *AI Risk Management Framework (AI RMF 1.0)*, 2023.;Barocas, Hardt, and Narayanan, *Fairness and Machine Learning* (online book), selected sections.;Knaflic, *Storytelling with Data*, selected chapters on communicating results.